AMERICAN INTERNATIONAL UNIVERSITY BANGLADESH (AIUB)

FACULTY OF SCIENCE & TECHNOLOGY

Course Title

INTRODUCTION TO DATABASE (2108)

Semester: Spring 23-24

Section: [X]

TITLE

Criminal Face Detection System

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Introduction

This project involves the development of a criminal face detection database management system, which is constructed using Oracle 10g and a software application called Canva. The aim of this project is to enhance global security by utilizing image recognition technology to scan the criminal history of specific individuals, thereby ensuring safer and more secure environments for people. This project holds significant importance in the tech sector as it eliminates the need for manual searches and background checks of individuals, replacing it with an automated, unmanned, and more efficient security search. For instance, security gates at airports are manned by security officers who manually scrutinize an individual's background, gaining a rough idea by interrogating them. However, by implementing this project, not only can we retrieve the entire history of a specific person simply by using their facial image, but we can also ensure that no criminals can forge their IDs or outsmart the security officers and escape. For the primary target consumers of this project, it will be highly beneficial for security forces worldwide. Their work will become more efficient by targeting ordinary civilians and scanning their facial images to check for any criminal records. As for potential secondary target consumers, financial institutions like banks and educational institutions such as colleges or universities could utilize the system to enhance security, prevent fraudulent activities, and ensure a safer environment by identifying potential threats among students and staff.

Case Study / Scenario

StudentID1: 23-50531-1 Name: AKIB HASAN PYIL	StudentID3: 23-50515-1 Name: MOST. SUMAIYA HABIBA NISHAT
StudentID2: 23-50559-1 Name: NAZAT E ROSE RHYTHM	StudentID4: 23-50511-1 Name: SANIA AKTER
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CO2: Understand the fundamental concepts underlying database systems and gain hands-on experience with ER diagram Case study	
PO-c2: Develop process for complex computer science and engineering problems considering cultural and societal factors.	Marks

CASE STUDY 1:

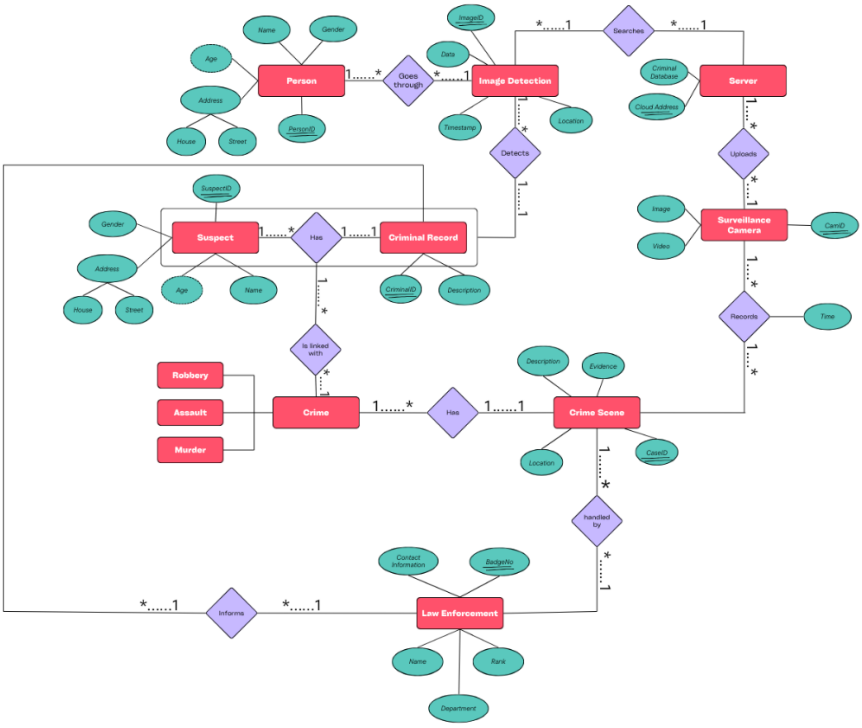
In a world where crime rate is at an all-time high, to maximize securities and key checkpoints, a face detection system has been applied that will scan the faces of individuals and check database to see if they have a criminal past or not. At first, A person goes through an image detection, the person has Name, Gender, a unique ID, Age and Address that will contain information about House number and Street name. Image Detection will store a unique Image ID, a date of when it was scanned, and timestamp, as well as the location. Image Detection will detect if the person is a Suspect and has Criminal Record. Criminal Record will have a unique criminal ID and description. Suspect will have Suspect ID, Age and Name, Gender, and address that will contain House and Street. A Suspect with a Criminal record will be linked with a Crime, there are three types of crime, Robbery, Assault and Murder. A crime will have a Crime Scene. Crime scene will have a unique Case ID, Description, Evidence, and location. On the other hand, Image Detection will search the Server which will host the Criminal Database and will have a Cloud Address. A Surveillance Camera, which will have a unique Cam Id and image and video attributes, will record the Crime Scene and upload it to the server. Once the image Detection detects a criminal past, it will inform the law enforcement, law enforcement will have badge No., Contact information, Name, Rank and Department and they will handle the Crime Scene.

CASE STUDY 2:

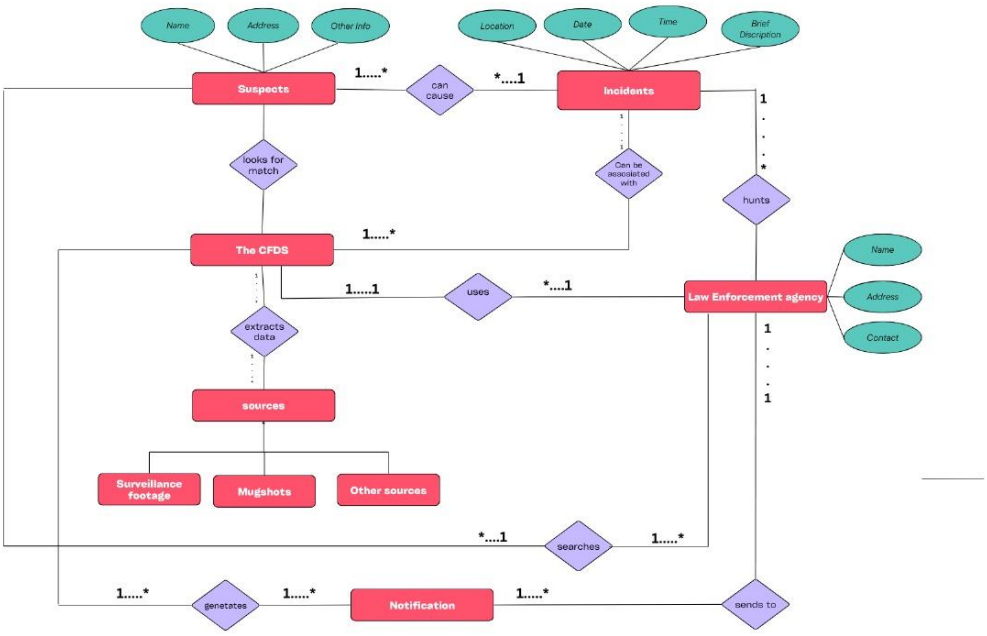
To address the criminal detecting challenge, Govt. have decided to implement a Criminal Face Detection System to enhance their investigative capabilities. **Suspects** are individuals who are potentially involved in criminal activities. They are identified by their name, address, and any other relevant personal information. Each suspect is associated with multiple incidents and is associated with multiple law enforcement agencies. **The CFDS** has a database of facial features extracted from surveillance footage, mugshots, and other sources. Each entry in the database corresponds to one suspect and may be associated with multiple incidents. **Incidents** represent criminal activities captured by surveillance cameras or reported by witnesses. Each incident is documented with details such as location, date, time, and a brief description of the event. Each incident is linked to one or more suspects and is associated with one or more law enforcement agencies. Multiple **law enforcement agencies** collaborate in using the CFDS. Each agency is identified by its name, address, and contact information. Each agency can be associated with multiple suspects and incidents. Upon identifying a match or significant event, the CFDS sends **notifications** to relevant law enforcement agencies, providing them with actionable intelligence to pursue investigations further. Each notification is generated for one or more incidents and is sent to one or more law enforcement agencies.

ER Diagram

The ER Diagram below is according to CASE STUDY 1.



The ER Diagram below is according to CASE STUDY 2:



Normalization

Goes through Relation,

UNF: Name, Age, Gender, Address, House , Street , PersonID, ImageID, Date, Timestamp, Location

1NF: Name, Age, Gender, Address, House , Street , PersonID, ImageID, Date, Timestamp, Location

2NF: 1. PersonID(pk), Name, Age, Gender, House , Street

2. ImageID(pk), Date, Timestamp, Location

3. PersonID(pk), ImageID(fk)

3NF: 1. PersonID(pk), Name, Age, Gender, Street

2. ImageID(pk), Date, Timestamp, Location

3. PersonID(pk), ImageID(fk)

Searches Relation,

UNF: ImageID, Date, Timestamp, Location, Criminal Database, Cloud Address

1NF: ImageID, Date, Timestamp, Location, Criminal Database, Cloud Address

2NF: 1. ImageID(pk), Date, Timestamp, Location

2. Cloud Address(pk), Criminal Database

3. ImageID(pk), Cloud Address(fk)

3NF: 1. ImageID(pk), Date, Timestamp, Location

2. Cloud Address(pk), Criminal Database

3. ImageID(pk), Cloud Address(fk)

Detects Relation,

UNF: ImageID, Date, Timestamp, Location, CriminalID, Description, SuspectID, Name, Age, Gender, Address, House, Street

1NF: ImageID, Date, Timestamp, Location, CriminalID, Description, SuspectID, Name, Age, Gender, Address, House, Street

2NF: 1. ImageID(pk), Date, Timestamp, Location

2. CriminalID(pk), Description, SuspectID(pk), Name, Age, Gender, House, Street

3. ImageID(pk), CriminalID(fk), SuspectID(fk)

3NF: 1. ImageID(pk), Date, Timestamp, Location

2. CriminalID(pk), Description, SuspectID(pk), Name, Age, Gender, Street

3. ImageID(pk), CriminalID(fk), SuspectID(fk)

Has Relation,

UNF: CriminalID, Description, SuspectID, Name, Age, Gender, Address, House, Street

1NF: CriminalID, Description, SuspectID, Name, Age, Gender, Address, House, Street

2NF: 1. CriminalID(pk), Description

2. SuspectID(pk), Name, Age, Gender, House, Street

3. CriminalID(pk), SuspectID(fk)

3NF: 1. CriminalID(pk), Description

2. SuspectID(pk), Name, Age, Gender, Street

3. CriminalID(pk), SuspectID(fk)

Is linked with Relation,

UNF: CriminalID, Description, SuspectID, Name, Age, Gender, Address, House, Street

1NF: CriminalID, Description, SuspectID, Name, Age, Gender, Address, House, Street

2NF: 1. CriminalID(pk), Description

2. SuspectID(pk), Name, Age, Gender, House, Street

3. CriminalID(pk), SuspectID(pk)

3NF: 1. CriminalID(pk), Description

2. SuspectID(pk), Name, Age, Gender, Street

3. CriminalID(pk), SuspectID(pk)

Uploads Relation,

UNF: Criminal Database, Cloud Address, CamID, Image, Video

1NF: Criminal Database, Cloud Address, CamID, Image, Video

2NF: 1. Criminal Database, Cloud Address (pk)

2. CamID(pk), Image, Video

3. CamID(pk), Cloud Address (fk)

3NF: 1. Criminal Database, Cloud Address (pk)

2. CamID(pk), Image, Video

3. CamID(pk), Cloud Address (fk)

Records Relation,

UNF: CamID, Image, Video, CaseID, Location, Description, Evidence

1NF: CamID, Image, Video, CaseID, Location, Description, Evidence

2NF: 1. CamID(pk), Image, Video

2. CaseID(pk), Location, Description, Evidence
3. CamID(pk), CaseID(fk),

3NF: 1. CamID(pk), Image, Video

2. CaseID(pk), Location, Description, Evidence
3. CamID(pk), CaseID(fk),

Has Relation,

UNF: CaseID, Location, Description, Evidence

1NF: CaseID, Location, Description, Evidence

2NF: 1. CaseID(pk), Location, Description, Evidence

2. CaseID(pk)

3NF: 1. CaseID(pk), Location, Description, Evidence

2. CaseID(pk)

Informs Relation,

UNF: CriminalID, Description, BadgeNO, Name, Rank, Contact Info, Department

1NF: CriminalID, Description, BadgeNO, Name, Rank, Contact Info, Department

2NF: 1. CriminalID(pk), Description

2. BadgeNO(pk), Name, Rank, Contact Info, Department
3. CriminalID(pk), BadgeNO(fk)

3NF: 1. CriminalID(pk), Description

2. BadgeNO(pk), Name, Rank, Contact Info, Department
3. CriminalID(pk), BadgeNO(fk)

Handled By Relation,

UNF: CaseID, Location , Description , Evidence, BadgeNO, Name, Rank, Contact Info, Department

1NF: CaseID, Location , Description , Evidence, BadgeNO, Name, Rank, Contact Info, Department

2NF: 1. CaseID(pk), Location , Description , Evidence

2 BadgeNO(pk), Name, Rank, Contact Info, Department

3. CaseID(pk), BadgeNO(fk)

3NF: 1. CaseID(pk), Location , Description , Evidence

2 BadgeNO(pk), Name, Rank, Contact Info, Department

3. CaseID(pk), BadgeNO(fk)

Finalization

1. PersonID(pk), Name, Age, Gender, Street, ImageID(fk)
2. ImageID(pk), Date, Timestamp, Location, cloud Address(fk), CriminalID(fk), SuspectID(fk)
3. Cloud Address(pk), Criminal Database
4. SuspectID(pk), Name, Age, Gender, Street, CriminalID(fk),
5. CamID(pk), Image, Video, Cloud Address (fk), CaseID(fk)
6. CaseID(pk), Location, Description, Evidence, BadgeNO(fk)
7. CriminalID(pk), Description, BadgeNO(fk)
8. BadgeNO(pk), Name, Rank, Contact Info, Department

Table Creation (DDL Operations)

StudentID1: 23-50531-1 Name: AKIB HASAN PYIL	StudentID3: 23-50515-1 Name: MOST. SUMAIYA HABIBA NISHAT
StudentID2: 23-50559-1 Name: NAZAT E ROSE RHYTHM	StudentID4: 23-50511-1 Name: SANIA AKTER
StudentID5: 23-51341-1 Name: SHAHADAT HOSSAIN	
CO4: Creating DML, DDL using Oracle and connection with ODBC/JDBC for existing JAVA application	
PO-e-2: Use modern engineering and IT tools for prediction and modeling of complex computer science and engineering problem	Marks

TABLE 1:

Home > SQL > SQL Commands

Autocommit Display 10

```
CREATE TABLE PersonInfo(
    PersonID NUMBER(10) PRIMARY KEY,
    Name VARCHAR2(256) ,
    Age NUMBER(5),
    Gender VARCHAR2(10) ,
    Street VARCHAR2(20) ,
    ImageID NUMBER(10),
    FOREIGN KEY (ImageID) REFERENCES ImageDetection(ImageID)
);
```

DESCRIPTION:

Object Type **TABLE** Object **PERSONINFO**

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
PERSONINFO	PERSONID	Number	-	10	0	1	-	-	-
	NAME	Varchar2	256	-	-	-	✓	-	-
	AGE	Number	-	5	0	-	✓	-	-
	GENDER	Varchar2	10	-	-	-	✓	-	-
	STREET	Varchar2	20	-	-	-	✓	-	-
	IMAGEID	Number	-	10	0	-	✓	-	-

1 - 6

TABLE 2:

Home > SQL > SQL Commands

Autocommit Display 10

```
CREATE TABLE ImageDetection (
    ImageID NUMBER(10) PRIMARY KEY NOT NULL,
    Crime_Date DATE,
    Crime_Timestamp TIMESTAMP,
    CloudAddress VARCHAR2(256) NOT NULL,
    CriminalID NUMBER(10),
    SuspectID NUMBER(10),
    FOREIGN KEY (CriminalID) REFERENCES Criminal(CriminalID),
    FOREIGN KEY (CloudAddress) REFERENCES Database(CloudAddress),
    FOREIGN KEY (SuspectID) REFERENCES Suspect(SuspectID)
);
```

DESCRIPTION:

Object Type **TABLE** Object **IMAGEDETECTION**

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
IMAGEDETECTION	IMAGEID	Number	-	10	0	1	-	-	-
	CRIME_DATE	Date	7	-	-	-	✓	-	-
	CRIME_TIMESTAMP	Timestamp(6)	11	-	6	-	✓	-	-
	CLOUDADDRESS	Varchar2	256	-	-	-	-	-	-
	CRIMINALID	Number	-	10	0	-	✓	-	-
	SUSPECTID	Number	-	10	0	-	✓	-	-

1 - 6

TABLE 3:

Home > SQL > SQL Commands

Autocommit Display 10

```
CREATE TABLE Database (
    CloudAddress VARCHAR2(256) NOT NULL PRIMARY KEY,
    CriminalDatabase VARCHAR2(256) NOT NULL
);
```

DESCRIPTION:

Object Type **TABLE** Object **DATABASE**

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
DATABASE	CLOUDADDRESS	Varchar2	256	-	-	1	-	-	-
	CRIMINALDATABASE	Varchar2	256	-	-	-	-	-	-

1 - 2

TABLE 4:

Home > SQL > SQL Commands

Autocommit Display 10

```
CREATE TABLE Suspect
(
    SuspectID NUMBER (10) PRIMARY KEY NOT NULL,
    Name VARCHAR2(256) NOT NULL ,
    Age NUMBER(5),
    Gender VARCHAR2(10) NOT NULL,
    Street VARCHAR2(20) NOT NULL,
    CriminalID NUMBER(10) NOT NULL,
    FOREIGN KEY (CriminalID) REFERENCES Criminal(CriminalID)
);
```

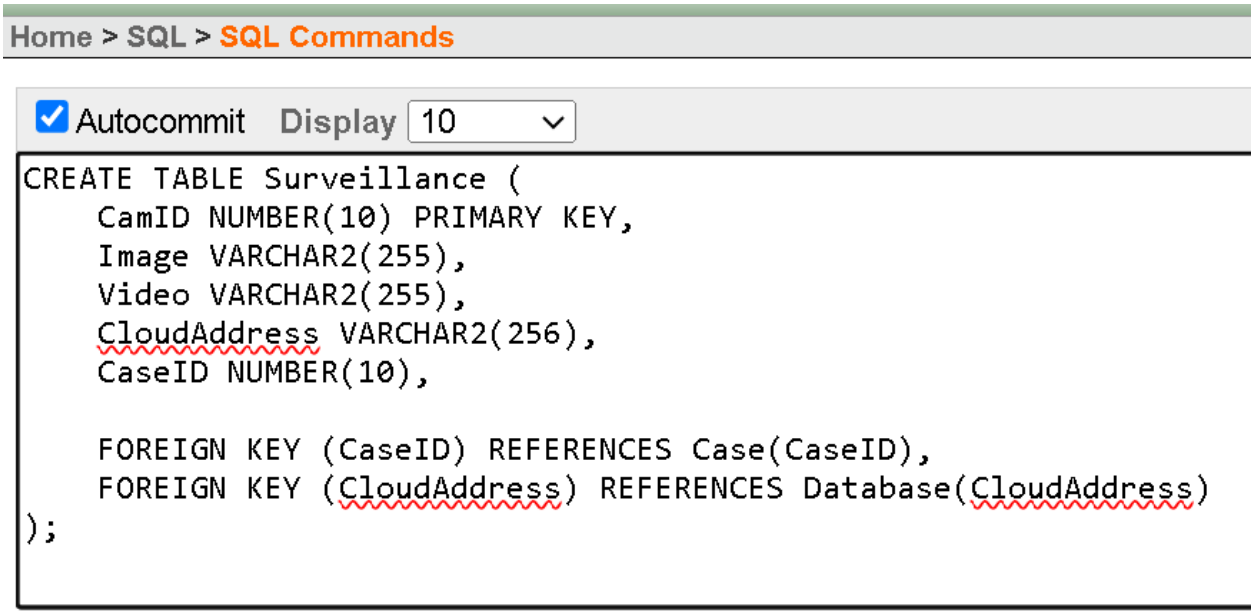
DESCRIPTION:

Object Type **TABLE** Object **SUSPECT**

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
SUSPECT	SUSPECTID	Number	-	10	0	1	-	-	-
	NAME	Varchar2	256	-	-	-	-	-	-
	AGE	Number	-	5	0	-	✓	-	-
	GENDER	Varchar2	10	-	-	-	-	-	-
	STREET	Varchar2	20	-	-	-	-	-	-
	CRIMINALID	Number	-	10	0	-	-	-	-

1 - 6

TABLE 5:



DESCRIPTION:

Object Type **TABLE** Object **SURVEILLANCE**

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
SURVEILLANCE	CAMID	Number	-	10	0	1	-	-	-
	IMAGE	Varchar2	255	-	-	-	✓	-	-
	VIDEO	Varchar2	255	-	-	-	✓	-	-
	CLOUDADDRESS	Varchar2	256	-	-	-	✓	-	-
	CASEID	Number	-	10	0	-	✓	-	-

1 - 5

TABLE 6:

Home > SQL > SQL Commands

Autocommit Display 10

```
CREATE TABLE Case (
    CaseID number (10)  NOT NULL PRIMARY KEY,
    Location VARCHAR(255),
    Description varchar(255),
    Evidence varchar(255),
    BadgeNO NUMBER(10) NOT NULL,

    FOREIGN KEY (BadgeNO) REFERENCES LawEnforcement(BadgeNo)
);
```

DESCRIPTION:

Object Type **TABLE** Object **CASE**

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
CASE	CASEID	Number	-	10	0	1	-	-	-
	LOCATION	Varchar2	255	-	-	-	✓	-	-
	DESCRIPTION	Varchar2	255	-	-	-	✓	-	-
	EVIDENCE	Varchar2	255	-	-	-	✓	-	-
	BADGENO	Number	-	10	0	-	-	-	-

1 - 5

TABLE 7:

Home > SQL > SQL Commands

Autocommit Display 10

```
CREATE TABLE Criminal (
    CriminalID NUMBER(10) PRIMARY KEY NOT NULL,
    Description VARCHAR2(255) NOT NULL,
    BadgeNo NUMBER(10),
    CONSTRAINT FK_BadgeNo
    FOREIGN KEY (BadgeNo) REFERENCES LawEnforcement(BadgeNo)
);
```

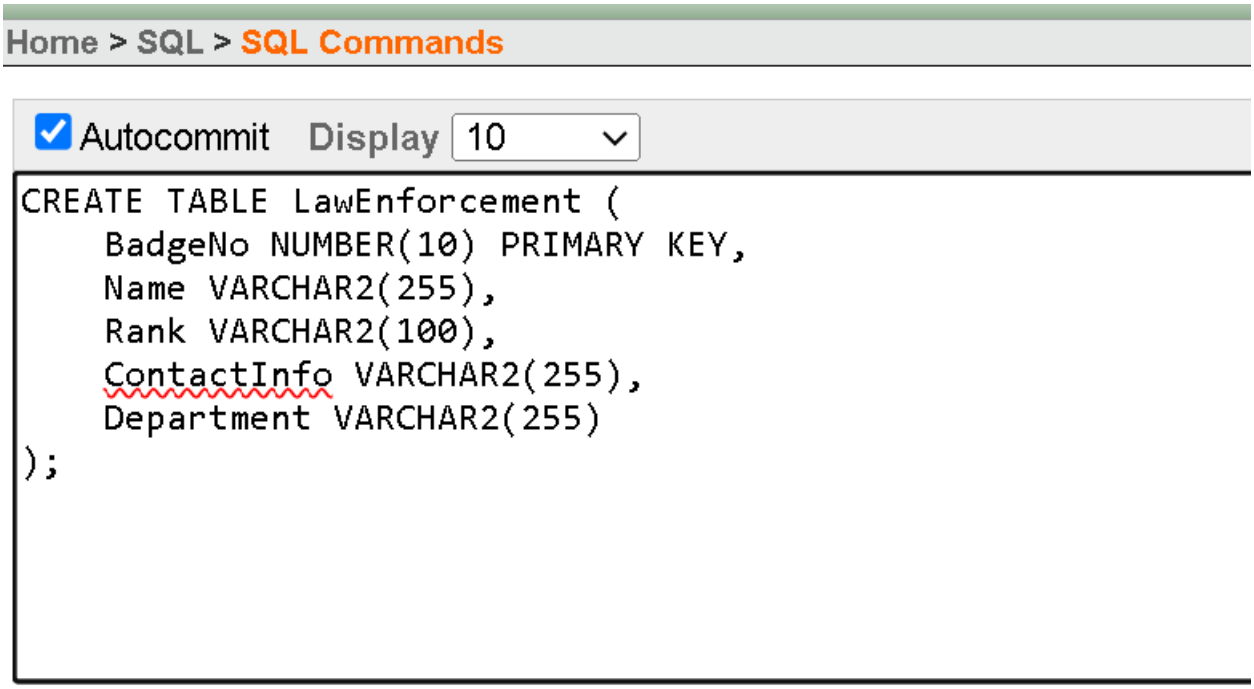
DESCRIPTION:

Object Type **TABLE** Object **CRIMINAL**

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
CRIMINAL	CRIMINALID	Number	-	10	0	1	-	-	-
	DESCRIPTION	Varchar2	255	-	-	-	-	-	-
	BADGENO	Number	-	10	0	-	✓	-	-

1 - 3

TABLE 8:



DESCRIPTION:

Object Type **TABLE** Object **LAWENFORCEMENT**

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
LAWENFORCEMENT	BADGENO	Number	-	10	0	1	-	-	-
	NAME	Varchar2	255	-	-	-	✓	-	-
	RANK	Varchar2	100	-	-	-	✓	-	-
	CONTACTINFO	Varchar2	255	-	-	-	✓	-	-
	DEPARTMENT	Varchar2	255	-	-	-	✓	-	-

1 - 5

Inserted Values in the tables

1. **Personinfo Table:**

PERSONID	NAME	AGE	GENDER	STREET	IMAGEID
1	John Doe	30	Male	123 Main St	1
2	Jane Smith	25	Female	456 Elm St	2
3	Jeni dev	30	Female	124 lin St	3
4	Smith Doe	40	Male	145 St	4
5	Doe Clark	48	Male	987 street	5
6	Blake	50	Male	345 lane st	6
7	Michel	37	Male	234 Main St	7
8	Jenifer	32	Female	789 Main St	8
9	Dev	30	Male	lane 23	9
10	Daniel	55	Male	1 Main St	10

2. **ImageDetection Table:**

IMAGEID	CRIME_DATE	CRIME_TIMESTAMP	CLOUDADDRESS	CRIMINALID	SUSPECTID
1	18-APR-24	18-APR-24 08.48.19.496000 AM	cloud1	-	1
2	18-APR-24	18-APR-24 08.48.22.576000 AM	cloud2	2	2
3	18-APR-24	18-APR-24 08.48.25.461000 AM	cloud3	3	3
4	18-APR-24	18-APR-24 08.48.28.446000 AM	cloud4	4	4
5	18-APR-24	18-APR-24 08.48.31.335000 AM	cloud5	5	5
6	18-APR-24	18-APR-24 08.48.34.544000 AM	cloud6	6	6
7	18-APR-24	18-APR-24 08.48.37.450000 AM	cloud7	7	7
8	18-APR-24	18-APR-24 08.48.40.246000 AM	cloud8	8	8
9	18-APR-24	18-APR-24 08.48.43.063000 AM	cloud9	9	9
10	18-APR-24	18-APR-24 08.48.45.617000 AM	cloud10	10	10

3. **Database Table:**

CLOUDADDRESS	CRIMINALDATABASE
cloud1	criminalDB1
cloud2	criminalDB2
cloud3	criminalDB3
cloud4	criminalDB4
cloud5	criminalDB5
cloud6	criminalDB6
cloud7	criminalDB7
cloud8	criminalDB8
cloud9	criminalDB9
cloud10	criminalDB10

4. **Suspect Table:**

SUSPECTID	NAME	AGE	GENDER	STREET	CRIMINALID
1	John Doe	30	Male	123 Main St	1
2	Jane Smith	25	Female	456 Elm St	2
3	Alice Johnson	35	Female	789 Oak St	3
4	Bob Williams	40	Male	101 Maple Ave	4
5	Eve Brown	28	Female	202 Pine St	5
6	Michael Lee	45	Male	303 Cedar Rd	6
7	Sophia Garcia	22	Female	404 Birch Ln	7
8	Daniel Martinez	33	Male	505 Oakwood Dr	8
9	Olivia Rodriguez	27	Female	606 Elmwood Ave	9
10	Liam Smith	32	Male	707 Maplewood Blvd	10

5. **Surveillance Table:**

CAMID	IMAGE	VIDEO	CLOUDADDRESS	CASEID
1	image1.jpg	video1.mp4	cloud1	1001
2	image2.jpg	video2.mp4	cloud2	1002
3	image3.jpg	video3.mp4	cloud3	1003
4	image4.jpg	video4.mp4	cloud4	1004
5	image5.jpg	video5.mp4	cloud5	1005
6	image6.jpg	video6.mp4	cloud6	1006
7	image7.jpg	video7.mp4	cloud7	1007
8	image8.jpg	video8.mp4	cloud8	1008
9	image9.jpg	video9.mp4	cloud9	1009
10	image10.jpg	video10.mp4	cloud10	1010

6. **Case Table:**

CASEID	LOCATION	DESCRIPTION	EVIDENCE	BADGENO
1006	Location 6	Description 6	Evidence 6	1234567895
1007	Location 7	Description 7	Evidence 7	1234567896
1008	Location 8	Description 8	Evidence 8	1234567897
1009	Location 9	Description 9	Evidence 9	1234567898
1010	Location 10	Description 10	Evidence 10	1234567899
1001	Location 1	Description 1	Evidence 1	1234567890
1002	Location 2	Description 2	Evidence 2	1234567891
1003	Location 3	Description 3	Evidence 3	1234567892
1004	Location 4	Description 4	Evidence 4	1234567893
1005	Location 5	Description 5	Evidence 5	1234567894

7. **Criminal Table:**

CRIMINALID	DESCRIPTION	BADGENO
1	Description 1	1234567890
2	Description 2	1234567891
3	Description 3	1234567892
4	Description 4	1234567893
5	Description 5	1234567894
6	Description 6	1234567895
7	Description 7	1234567896
8	Description 8	1234567897
9	Description 9	1234567898
10	Description 10	1234567899

8. **LawEnforcement Table:**

BADGENO	NAME	RANK	CONTACTINFO	DEPARTMENT
1234567890	Officer Smith	Sergeant	smith@example.com	Patrol Division
1234567891	Officer Johnson	Detective	johnson@example.com	Investigations Division
1234567892	Officer Williams	Lieutenant	williams@example.com	Special Operations Unit
1234567893	Officer Brown	Captain	brown@example.com	Administrative Division
1234567894	Officer Jones	Officer	jones@example.com	Traffic Division
1234567895	Officer Davis	Sergeant	davis@example.com	K-9 Unit
1234567896	Officer Miller	Detective	miller@example.com	Narcotics Unit
1234567897	Officer Wilson	Lieutenant	wilson@example.com	Community Policing Unit
1234567898	Officer Moore	Captain	moore@example.com	Special Investigations Unit
1234567899	Officer Taylor	Officer	taylor@example.com	Cybercrime Unit

Query Test in DB

a) SIMPLE QUERIES

SELECT * FROM PersonInfo WHERE Gender = 'Male';

SELECT * FROM PersonInfo WHERE Age > 10;

SELECT * FROM PersonInfo WHERE Street = '145 st';

SELECT * FROM PersonInfo WHERE PersonID = 1;

PERSONID	NAME	AGE	GENDER	STREET	IMAGEID
1	John Doe	30	Male	123 Main St	1

b) Query with a single row function

SELECT LOWER(Image) AS Lowercase_Image FROM Surveillance;

SELECT Name, UPPER(Gender) AS Gender FROM Suspect;

NAME	GENDER
John Doe	MALE
Jane Smith	FEMALE
Alice Johnson	FEMALE
Bob Williams	MALE
Eve Brown	FEMALE
Michael Lee	MALE
Sophia Garcia	FEMALE
Daniel Martinez	MALE
Olivia Rodriguez	FEMALE
Liam Smith	MALE

c) Query with a Multiple row function/ aggregate function

SELECT Gender, COUNT(*) AS Gender_Count FROM PersonInfo GROUP BY Gender;

SELECT MAX(Age) AS Max_Age, MIN(Age) AS Min_Age FROM PersonInfo;

MAX_AGE	MIN_AGE
55	25

d) Single row subquery and multiple row subquery

SELECT Name, Age FROM PersonInfo WHERE Age = (SELECT MAX(Age) FROM PersonInfo);

SELECT Name, Age FROM PersonInfo WHERE Age = (SELECT MIN(Age) FROM PersonInfo);

NAME	AGE
Jane Smith	25

SELECT Name FROM Suspect WHERE SuspectID IN (SELECT SuspectID FROM ImageDetection);

SELECT Name FROM Suspect WHERE Age > (SELECT AVG(Age) FROM Suspect);

NAME
Alice Johnson
Bob Williams
Michael Lee
Daniel Martinez
Liam Smith

e) 4 kinds of joining

INNER JOIN

SELECT PersonInfo.Name, ImageDetection.Crime_Date FROM PersonInfo INNER JOIN ImageDetection ON PersonInfo.ImageID = ImageDetection.ImageID;

NAME	CRIME_DATE
John Doe	18-APR-24
Jane Smith	18-APR-24
Jeni dev	18-APR-24
Smith Doe	18-APR-24
Doe Clark	18-APR-24
Blake	18-APR-24
Michel	18-APR-24
Jenifer	18-APR-24
Dev	18-APR-24
Daniel	18-APR-24

LEFT JOIN

SELECT PersonInfo.Name, ImageDetection.CloudAddress FROM PersonInfo LEFT JOIN ImageDetection ON PersonInfo.ImageID = ImageDetection.ImageID;

NAME	CLOUDADDRESS
John Doe	cloud1
Jane Smith	cloud2
Jeni dev	cloud3
Smith Doe	cloud4
Doe Clark	cloud5
Blake	cloud6
Michel	cloud7
Jenifer	cloud8
Dev	cloud9
Daniel	cloud10

RIGHT JOIN

SELECT ImageDetection.CloudAddress, PersonInfo.Name FROM PersonInfo RIGHT JOIN ImageDetection ON PersonInfo.ImageID = ImageDetection.ImageID;

CLOUDADDRESS	NAME
cloud1	John Doe
cloud2	Jane Smith
cloud3	Jeni dev
cloud4	Smith Doe
cloud5	Doe Clark
cloud6	Blake
cloud7	Michel
cloud8	Jenifer
cloud9	Dev
cloud10	Daniel

FULL OUTER JOIN

SELECT PersonInfo.Name, ImageDetection.CloudAddress FROM PersonInfo

FULL OUTER JOIN ImageDetection ON PersonInfo.ImageID = ImageDetection.ImageID;

NAME	CLOUDADDRESS
John Doe	cloud1
Jane Smith	cloud2
Jeni dev	cloud3
Smith Doe	cloud4
Doe Clark	cloud5
Blake	cloud6
Michel	cloud7
Jenifer	cloud8
Dev	cloud9
Daniel	cloud10

f) Simple view

```
CREATE VIEW Simple_Person_View AS

SELECT PersonID, Name, Age, Gender, Street

FROM scott.PersonInfo;
```

g) Complex view

```
CREATE VIEW Complex_Person_Surveillance_View AS
SELECT pi.PersonID, pi.Name AS PersonName, pi.Age, pi.Gender, pi.Street,
    id.Crime_Date, id.Crime_Timestamp, s.CamID, s.Image, s.Video,
    s.CloudAddress AS SurveillanceCloud, c.Location AS CaseLocation, c.Description AS
CaseDescription
FROM scott.PersonInfo pi
JOIN scott.ImageDetection id ON pi.ImageID = id.ImageID
JOIN scott.Surveillance s ON id.CloudAddress = s.CloudAddress
JOIN scott.Case c ON s.CaseID = c.CaseID;
```

Description of a Successful DB connection

Setting up Criminal Face Detection System Database and Java Connection

The initial step in launching the Criminal Face Detection System involved the installation of necessary software and JAR files. This encompassed configuring the **VSCODE IDE**, installing the **Xampp Server**, and acquiring the **mysql-connector-java-8.0.28.jar file**.

With the requisite software installed, we proceeded to launch the Xampp control panel, initiating the Apache and MySQL modules. Accessing phpMyAdmin, I commenced the creation of the database "Criminal Face Detection System," followed by the establishment of 5 tables "Personinfo" ,“Case”, “Criminal”, “LawEnforcements”,”Suspect”, . Then we put some values in these.

Upon completing the database and table setup, we proceeded to intelliJ where we established a project. Here, we incorporated the mysqlconnector-java-8.0.28.jar as a library. Within the main file, we crafted code to establish a connection with the Xampp-created database.

We all 5 followed the same procedure as above.

Here are our Table Creation and DB connection Proofs:

1. Personinfo Table by SHAHADAT HOSSAIN(23-51341-1):

PersonID	Name	Age	Gender	Street	ImageID
1	John Doe	30	Male	123 Main St	1
2	Jane Smith	25	Female	456 Elm St	2
3	Jeni dev	30	Female	124 lin St	3
4	Smith Doe	40	Male	145 St	4
5	Doe Clark	48	Male	987 street	5
6	Blake	50	Male	345 lane st	6
7	Michel	37	Male	234 Main St	7
8	Jenifer	32	Female	789 Main St	8
9	Dev	30	Male	lane 23	9
10	Daniel	55	Male	1 Main St	10

Connection Code:



Output:



2. Case Table by NAZAT E ROSE RHYTHM (23-50559-1:

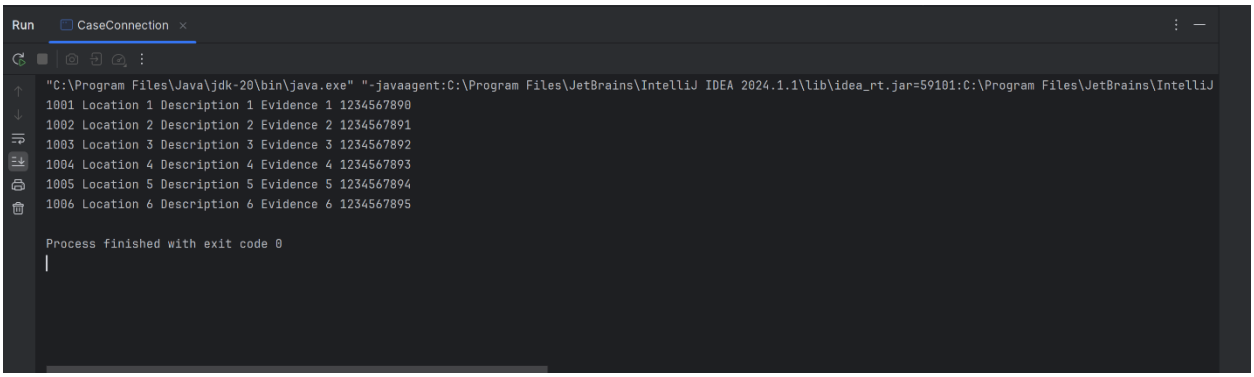
CaseID	Location	Description	Evidence	BadgeNO
1001	Location 1	Description 1	Evidence 1	1234567890
1002	Location 2	Description 2	Evidence 2	1234567891
1003	Location 3	Description 3	Evidence 3	1234567892
1004	Location 4	Description 4	Evidence 4	1234567893
1005	Location 5	Description 5	Evidence 5	1234567894
1006	Location 6	Description 6	Evidence 6	1234567895

Show all | Number of rows: 25 Filter rows: Search this table

Connection Code:



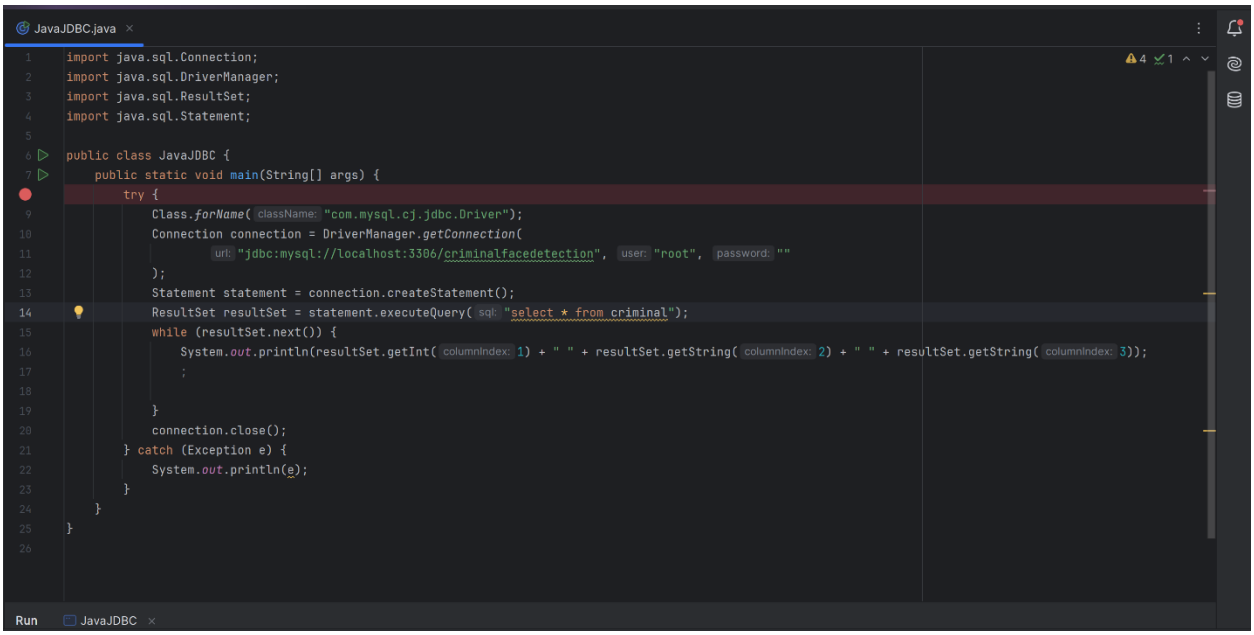
Output:



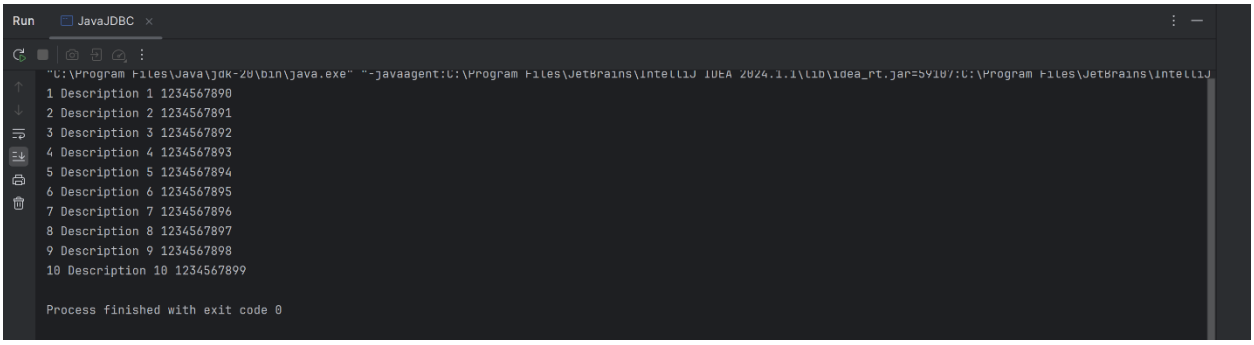
3. Criminal Table by SUMAIYA HABIBA NISHAT (23-50515-1):

criminalid	description	badgeno
1	Description 1	1234567890
2	Description 2	1234567891
3	Description 3	1234567892
4	Description 4	1234567893
5	Description 5	1234567894
6	Description 6	1234567895
7	Description 7	1234567896
8	Description 8	1234567897
9	Description 9	1234567898
10	Description 10	1234567899

Connection Code:



Output:



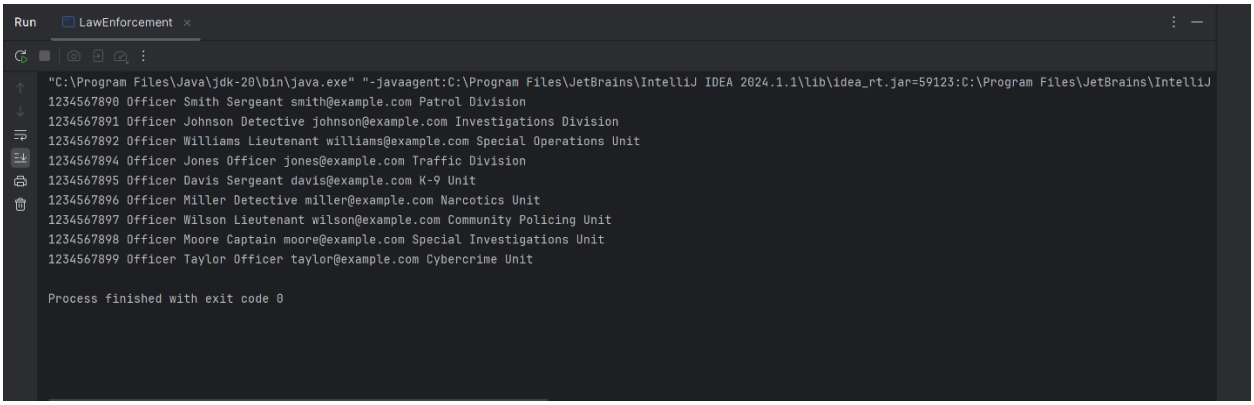
4. LawEnforcement Table by AKIB HASAN PYIL (23-50531-1):

BadgeNo	Name	Rank	ContactInfo	Department
1234567890	Officer Smith	Sergeant	smith@example.com	Patrol Division
1234567891	Officer Johnson	Detective	johnson@example.com	Investigations Division
1234567892	Officer Williams	Lieutenant	williams@example.com	Special Operations Unit
1234567894	Officer Jones	Officer	jones@example.com	Traffic Division
1234567895	Officer Davis	Sergeant	davis@example.com	K-9 Unit
1234567896	Officer Miller	Detective	miller@example.com	Narcotics Unit
1234567897	Officer Wilson	Lieutenant	wilson@example.com	Community Policing Unit
1234567898	Officer Moore	Captain	moore@example.com	Special Investigations Unit
1234567899	Officer Taylor	Officer	taylor@example.com	Cybercrime Unit

Connection Code:



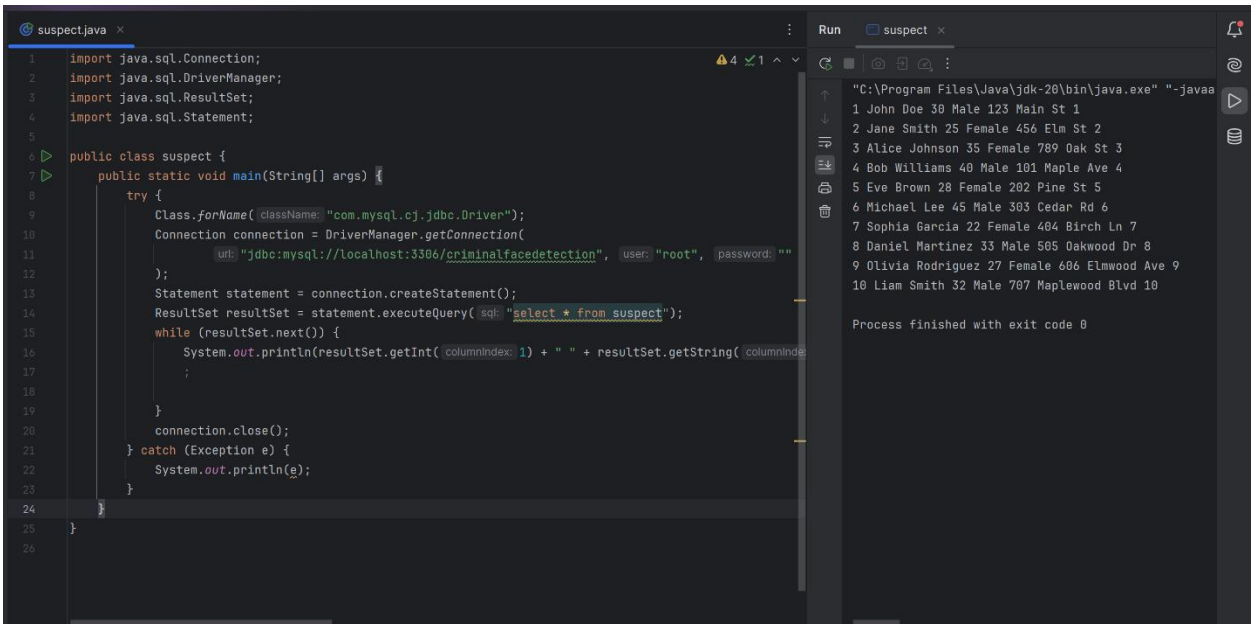
Output:



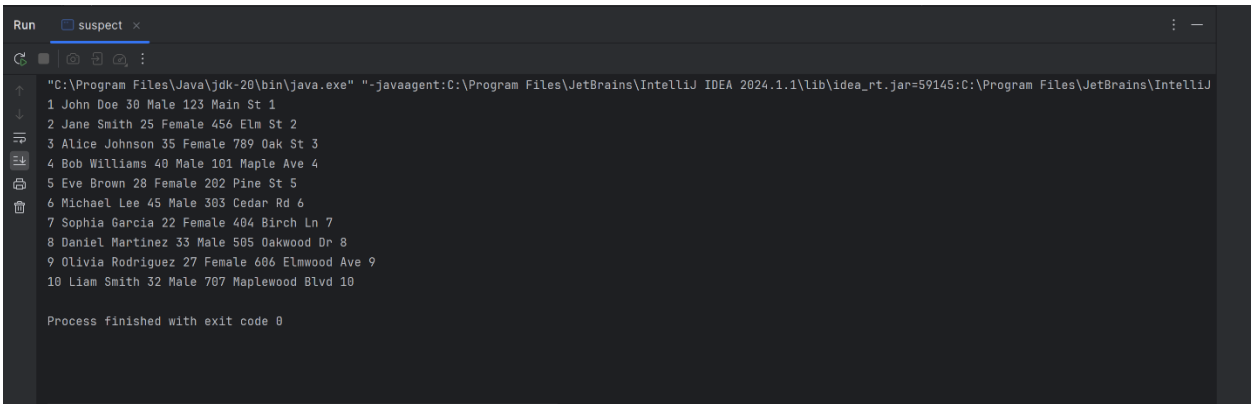
5. Suspect Table by SANIA AKTER (23-50511-1):

SuspectID	Name	Age	Gender	Street	CriminalID
1	John Doe	30	Male	123 Main St	1
2	Jane Smith	25	Female	456 Elm St	2
3	Alice Johnson	35	Female	789 Oak St	3
4	Bob Williams	40	Male	101 Maple Ave	4
5	Eve Brown	28	Female	202 Pine St	5
6	Michael Lee	45	Male	303 Cedar Rd	6
7	Sophia Garcia	22	Female	404 Birch Ln	7
8	Daniel Martinez	33	Male	505 Oakwood Dr	8
9	Olivia Rodriguez	27	Female	606 Elmwood Ave	9
10	Liam Smith	32	Male	707 Maplewood Blvd	10

Connection Code:



Output:



Conclusion

The criminal face detection database management system, developed using Oracle 10g and Canva, represents a significant advancement in global security technology. By leveraging image recognition technology, the system aims to revolutionize security protocols by automating the process of background checks and identification of individuals with criminal histories. Unlike traditional manual methods, which are prone to errors and manipulation, this system offers an efficient, unmanned, and foolproof solution for identifying potential threats.

The project holds immense importance in the tech sector as it eliminates the need for manual searches and background checks, thus enhancing efficiency and accuracy in security procedures. For primary target consumers such as security forces, the system offers a streamlined approach to identifying individuals with criminal records, thereby making their work more efficient and effective.

Additionally, secondary target consumers like financial institutions and educational establishments stand to benefit from the enhanced security measures provided by the system. Banks can prevent fraudulent activities, while colleges and universities can ensure a safer environment by identifying potential threats among students and staff.

Looking to the future, the project holds promising prospects for further development and expansion. Future work could involve refining the image recognition algorithms for greater accuracy, integrating additional security features, and extending the application to other sectors beyond law enforcement. Overall, the project represents a significant step forward in enhancing global security and safety.